

LandScan AI — Major Updates (3-Person Work Division)

181 Major Modules Divided Across 3 Engineers | Includes Backend & UI/UX Tasks

Team Workload Distribution Overview

Team Member	Core Role & Focus	Modules	Key UI/UX Deliverable
Person 1	OCR Processing & Image Preprocessing Specialist	40 Modules (Cat E, F, G)	OCR raw text viewer, polygon bounding box overlay & scan quality preview modal
Person 2	NLP Field Extractor & Land Rule Validator Lead	72 Modules (Cat H, I, J)	Color-coded confidence badges, rule failure inspection list & GIS area mismatch overlay
Person 3	Human Review Queue, Audit Trail & API Lead	69 Modules (Cat K, L, M, N, O, P, Q, S)	Dedicated /review queue page, side-by-side verification drawer & audit log details

■■■ PERSON 1: OCR Processing & Image Preprocessing Specialist (40 Modules)

Backend Tasks:

- Build OpenCV preprocessing engine (grayscale, binarization, contrast boost, deskewing) in ``app/services/preprocessing.py`` (E01–E10).
- Build PaddleOCR / Tesseract engine (``app/services/ocr_service.py``) for PDF rasterization, text line parsing, polygon bounding box generation & multilingual support (F01–F20).
- Build CER / WER OCR ground-truth evaluation script (G01–G10).

UI/UX Deliverables (Person 1):

- In-Browser Document OCR Raw Text & Polygon Overlay Inspection Drawer component.
- Document Scan Quality Preview & Preprocessing Toggle Modal in upload window.
- OCR Accuracy Analytics Dashboard Widget (Accuracy by Language & Image Quality).

■■■ PERSON 2: NLP Field Extractor & Land Rule Validator Lead (72 Modules)

Backend Tasks:

- Build Structured Land Field Extractor (``app/services/extractor.py``): Regex & NLP parsers for Khasra, Khata, Patta, Owner Name, Survey No, Area (ha/sq.m), Village, District, Dates & Multilingual Aliases (H01–H30).
- Build Field & Document Confidence Scoring Engine: Dynamic weighting (OCR score + Regex precision + Gazetteer lookup), score thresholds (I01–I15).
- Build Land Record Business Rule Engine (``app/services/land_validator.py``): Survey/Khasra format checks, Duplicate parcel detection, Joint ownership 100% share sum check, and GIS vs Document area mismatch calculator (J01–J30).

UI/UX Deliverables (Person 2):

- Color-Coded Field Confidence Badges (``Green ≥90%``, ``Yellow 70-89%``, ``Orange 40-69%``, ``Red <40%``) with tooltip explanations.
- Land Validation Error & Warning Rule Inspection List Panel in document detail view.

- GIS Map Area Mismatch & Validation Status Layer Toggle (`Green=Verified`, `Yellow=In Review`, `Red=Failed`) on Leaflet map.

■■■ PERSON 3: Human Review Queue, Audit Trail & API Lead (69 Modules)

Backend Tasks:

- Build Human Review Queue Workflow APIs (`/api/review`, approve, reject, field correction endpoints) and status transitions (K01–K24, P04–P10).
- Extend Cryptographic SHA-256 Audit Log (`app/audit.py`) for OCR completion, field extraction, validation failures, manual corrections & approval events (L03–L08).
- Relational DB Schemas (`DocumentOCRResult`, `ExtractedField`, `ValidationResult`, `ReviewQueue`) & automated test suite (Q04–Q08, S02–S05).

UI/UX Deliverables (Person 3):

- Dedicated Human Review Queue Page (`/review`) with searchable/filterable queue table.
- Side-by-Side Reviewer Interface Drawer: Left side original document scan preview, Right side editable extracted field form with Approve, Reject, Escalate buttons and Comment box.
- Reviewer Action History & Cryptographic Audit Log Detail Drawer.

Complete Module Assignment Table (181 Modules)

Itemized breakdown showing the exact assignment and UI/UX responsibility for every module:

ID	Module / Requirement	Assigned Engineer	Priority	UI/UX & Engineering Task
C08	Verifier can review records	Person 3	P0	Task: Review queue & verification flow missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
E01	Original image is preserved	Person 1	P1	Task: Preprocessing module missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
E02	Grayscale preprocessing exists	Person 1	P1	Task: Grayscale processing missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
E03	Image resizing exists	Person 1	P2	Task: Image resizing missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
E04	Denoising exists	Person 1	P1	Task: OpenCV denoising missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
E05	Contrast enhancement exists	Person 1	P1	Task: Contrast boost missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
E06	Thresholding/binarization exists	Person 1	P1	Task: Binarization missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
E07	Deskewing exists	Person 1	P2	Task: Deskewing missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
E08	Preprocessed image is saved	Person 1	P2	Task: Processed file storage missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
E09	Preprocessing can be enabled or disabled	Person 1	P2	Task: Toggle option missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
E10	Image-processing errors are handled	Person 1	P1	Task: Exception handling missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F01	PaddleOCR package is installed	Person 1	P0	Task: Package installation missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F02	PaddleOCR is imported correctly	Person 1	P0	Task: Import missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F03	OCR runs on an actual uploaded image	Person 1	P0	Task: OCR invocation missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F04	OCR runs on a PDF or PDF page	Person 1	P1	Task: PDF OCR pipeline missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F05	OCR output is not only hard-coded mock text	Person 1	P0	Task: Real OCR text generation missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal

F06	OCR raw text is saved	Person 1	P0	Task: Database column missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F07	OCR result is linked to document ID	Person 1	P0	Task: Database relationship missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F08	OCR confidence is captured	Person 1	P0	Task: Confidence metric missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F09	OCR text boxes or polygons are captured	Person 1	P1	Task: Bounding box storage missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F10	OCR language is configurable	Person 1	P1	Task: Language parameter missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F11	English OCR is tested	Person 1	P1	Task: Unit test missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F12	Hindi OCR is configured or tested	Person 1	P1	Task: Hindi OCR missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F13	Telugu OCR is configured or tested	Person 1	P1	Task: Telugu OCR missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F14	OCR processing time is recorded	Person 1	P2	Task: Execution timing missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F15	OCR result can be viewed in the UI	Person 1	P0	Task: UI viewer component missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F16	OCR failure is shown clearly	Person 1	P1	Task: Error UI missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F17	Multi-page documents are handled	Person 1	P2	Task: Multi-page loop missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F18	OCR result can be exported	Person 1	P2	Task: Export route missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F19	OCR model version is recorded	Person 1	P2	Task: Model metadata missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
F20	OCR output is reproducible	Person 1	P2	Task: Reproducibility verification missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
G01	OCR ground truth is loaded	Person 1	P2	Task: Ground truth dataset missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
G02	Actual OCR output is compared with ground truth	Person 1	P2	Task: Evaluation comparator missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
G03	Character Error Rate is calculated	Person 1	P2	Task: CER calculation missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal

G04	Word Error Rate is calculated	Person 1	P2	Task: WER calculation missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
G05	OCR confidence is reported	Person 1	P1	Task: Confidence dashboard widget missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
G06	OCR accuracy is reported by language	Person 1	P2	Task: Language analytics missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
G07	OCR accuracy is reported by image quality	Person 1	P2	Task: Quality analytics missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
G08	OCR processing time is reported	Person 1	P2	Task: Latency metrics missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
G09	Evaluation does not compare output with itself	Person 1	P2	Task: Evaluation comparator missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
G10	Evaluation report can be exported	Person 1	P2	Task: Report export missing UI/UX: OCR raw text viewer, bounding box overlay & scan quality preview modal
H01	Owner name is extracted	Person 2	P0	Task: Owner name parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H02	Survey number is extracted	Person 2	P0	Task: Survey number parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H03	Khasra number is extracted	Person 2	P0	Task: Khasra number parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H04	Khata number is extracted	Person 2	P0	Task: Khata number parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H05	Patta number is extracted	Person 2	P0	Task: Patta number parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H06	Area value is extracted	Person 2	P0	Task: Area value parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H07	Area unit is extracted	Person 2	P0	Task: Area unit parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H08	Area is converted to square metres	Person 2	P1	Task: Unit converter missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H09	Village is extracted	Person 2	P0	Task: Village parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H10	Tehsil/Mandal is extracted	Person 2	P0	Task: Tehsil parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H11	District is extracted	Person 2	P0	Task: District parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay

H12	State is extracted	Person 2	P0	Task: State parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H13	Land classification is extracted	Person 2	P1	Task: Classification parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H14	Land use is extracted	Person 2	P1	Task: Land use parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H15	Registration number is extracted	Person 2	P1	Task: Registration number parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H16	Registration date is extracted	Person 2	P1	Task: Registration date parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H17	Mutation number is extracted	Person 2	P1	Task: Mutation number parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H18	Mutation date is extracted	Person 2	P1	Task: Mutation date parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H19	Mutation status is extracted	Person 2	P1	Task: Mutation status parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H20	Encumbrance status is extracted	Person 2	P1	Task: Encumbrance status parser missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H21	Regex extraction works	Person 2	P0	Task: Regex patterns missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H22	Dictionary/location matching works	Person 2	P1	Task: Dictionary matcher missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H23	Fuzzy matching works	Person 2	P1	Task: Fuzzy matching missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H24	Multilingual field aliases exist	Person 2	P1	Task: Alias dictionary missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H25	Extracted value is separate from normalized value	Person 2	P1	Task: Dual field schema missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H26	Extraction source is recorded	Person 2	P1	Task: Source attribute missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H27	Missing fields are handled	Person 2	P1	Task: Missing field handler missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H28	Extraction output is saved	Person 2	P0	Task: DB table missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
H29	Extraction output is linked to document ID	Person 2	P0	Task: Foreign key missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay

H30	Extraction output is linked to record ID	Person 2	P0	Task: Foreign key missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I01	Field-level confidence exists	Person 2	P0	Task: Confidence algorithm missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I02	OCR confidence contributes to final confidence	Person 2	P0	Task: Confidence weighting missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I03	Extraction confidence contributes to final confidence	Person 2	P1	Task: Pattern scoring missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I04	Master-data match contributes to confidence	Person 2	P1	Task: Master-data validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I05	Validation result contributes to confidence	Person 2	P1	Task: Rule feedback scoring missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I06	Confidence thresholds are configured	Person 2	P0	Task: Threshold config missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I07	High-confidence fields are marked green	Person 2	P1	Task: Green badge missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I08	Medium-confidence fields are marked yellow	Person 2	P1	Task: Yellow badge missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I09	Low-confidence fields are marked red	Person 2	P1	Task: Red badge missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I10	Low-confidence fields are routed to review	Person 2	P0	Task: Routing trigger missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I11	Confidence is not randomly regenerated	Person 2	P1	Task: Deterministic scoring missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I12	Confidence explanation is visible	Person 2	P2	Task: Tooltip UI missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I13	Overall document confidence is calculated	Person 2	P1	Task: Composite score missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I14	Confidence is stored with extracted field	Person 2	P0	Task: Column missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
I15	Confidence is included in dashboard metrics	Person 2	P1	Task: Dashboard metric missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J04	Area-unit validation works	Person 2	P1	Task: Unit validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J05	Survey-number format validation works	Person 2	P1	Task: Regex validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay

J06	Khasra-number validation works	Person 2	P1	Task: Khasra validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J07	Owner-name validation works	Person 2	P1	Task: Name validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J08	Village validation works	Person 2	P1	Task: Village validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J09	District validation works	Person 2	P1	Task: District validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J10	District metadata consistency works	Person 2	P1	Task: Hierarchy validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J11	Village/tehsil consistency works	Person 2	P1	Task: Hierarchy validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J12	Duplicate parcel detection works	Person 2	P0	Task: Duplicate detector missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J13	Duplicate survey detection works	Person 2	P0	Task: Duplicate detector missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J14	Owner share validation works	Person 2	P1	Task: Share validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J15	Joint ownership share validation works	Person 2	P1	Task: Share sum validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J16	Registration/mutation date validation works	Person 2	P1	Task: Date validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J17	Registration number validation works	Person 2	P1	Task: Registration validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J18	Land classification validation works	Person 2	P1	Task: Classification validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J19	Ownership-type validation works	Person 2	P1	Task: Type validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J20	GIS parcel existence check works	Person 2	P1	Task: GIS validator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J21	GIS/text area mismatch check works	Person 2	P0	Task: Area comparator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J22	Low OCR confidence validation works	Person 2	P0	Task: Confidence rule missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J23	Validation status is generated	Person 2	P0	Task: Status generator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay

J24	Validation severity is generated	Person 2	P0	Task: Severity generator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J25	Validation message is generated	Person 2	P0	Task: Message generator missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J26	Validation result is saved	Person 2	P0	Task: DB table missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J27	Validation result is linked to document	Person 2	P0	Task: Foreign key missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J28	Validation result is linked to rule ID	Person 2	P0	Task: Column missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J29	High-severity issues require review	Person 2	P0	Task: Trigger missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
J30	Validation results are visible in UI	Person 2	P0	Task: UI panel missing UI/UX: Color-coded confidence badges, rule failure list & GIS area mismatch overlay
K01	Review queue page exists	Person 3	P0	Task: Review queue page missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K02	Low-confidence records enter review queue	Person 3	P0	Task: Queue populator missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K03	Validation-failure records enter review queue	Person 3	P0	Task: Queue populator missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K04	Review queue shows document ID	Person 3	P1	Task: Table UI missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K05	Review queue shows field name	Person 3	P1	Task: Table UI missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K06	Review queue shows confidence	Person 3	P1	Task: Table UI missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K07	Review queue shows issue type	Person 3	P1	Task: Table UI missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K08	Review queue shows priority	Person 3	P1	Task: Table UI missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K09	Original document is visible	Person 3	P0	Task: Viewer component missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K10	OCR text is visible	Person 3	P1	Task: Viewer component missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K11	Extracted fields are editable	Person 3	P0	Task: Form component missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail

K12	Reviewer can correct a field	Person 3	P0	Task: Action handler missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K13	Reviewer can add a comment	Person 3	P1	Task: Comment input missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K14	Reviewer can save correction	Person 3	P0	Task: Save API missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K15	Reviewer can approve record	Person 3	P0	Task: Approve API missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K16	Reviewer can reject record	Person 3	P0	Task: Reject API missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K17	Reviewer can escalate record	Person 3	P2	Task: Escalate API missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K18	Previous value is preserved	Person 3	P1	Task: History tracking missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K19	Corrected value is stored	Person 3	P0	Task: DB column missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K20	Record status changes after review	Person 3	P0	Task: Transition logic missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K21	Dashboard updates after review	Person 3	P1	Task: Trigger missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K22	Review status is stored	Person 3	P0	Task: Column missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K23	Reviewer ID is stored	Person 3	P1	Task: Column missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
K24	Review timestamp is stored	Person 3	P1	Task: Column missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
L03	OCR completion is logged	Person 3	P1	Task: Event logger missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
L04	Extraction is logged	Person 3	P1	Task: Event logger missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
L05	Validation is logged	Person 3	P1	Task: Event logger missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
L07	Approval is logged	Person 3	P1	Task: Event logger missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
L08	Rejection is logged	Person 3	P1	Task: Event logger missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail

M03	Processed documents metric works	Person 3	P1	Task: Metric card missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
M04	Verified documents metric works	Person 3	P1	Task: Metric card missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
M05	Pending review metric works	Person 3	P1	Task: Metric card missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
M06	Rejected documents metric works	Person 3	P1	Task: Metric card missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
M07	Average confidence metric works	Person 3	P1	Task: Metric card missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
M08	Validation issue metric works	Person 3	P1	Task: Metric card missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
M12	Language distribution chart works	Person 3	P2	Task: Chart missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
M13	Validation issue chart works	Person 3	P2	Task: Chart missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
N05	Verified parcel color works	Person 3	P1	Task: Verification status color palette missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
N06	Review parcel color works	Person 3	P1	Task: Review status color missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
N07	Error parcel color works	Person 3	P1	Task: Error status color missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
N08	Processing parcel color works	Person 3	P1	Task: Processing status color missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
N12	Parcel links to document	Person 3	P2	Task: Document link missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
N15	Land-use filter works	Person 3	P2	Task: Land-use filter missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
N17	GIS/text area mismatch is displayed	Person 3	P1	Task: Mismatch indicator missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
O02	OCR accuracy report exists	Person 3	P2	Task: Report endpoint missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
O03	Field extraction accuracy report exists	Person 3	P2	Task: Report endpoint missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
O04	Validation failure-rate report exists	Person 3	P2	Task: Report endpoint missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail

O05	Human-review rate report exists	Person 3	P2	Task: Report endpoint missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
O06	Correction-rate report exists	Person 3	P2	Task: Report endpoint missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
O07	Processing-time report exists	Person 3	P2	Task: Report endpoint missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
O08	Language-wise report exists	Person 3	P2	Task: Report endpoint missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
O09	Image-quality-wise report exists	Person 3	P2	Task: Report endpoint missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
P04	OCR-processing API exists	Person 3	P0	Task: API route missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
P05	Extraction API exists	Person 3	P0	Task: API route missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
P06	Validation API exists	Person 3	P0	Task: API route missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
P07	Review-queue API exists	Person 3	P0	Task: API route missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
P08	Approve API exists	Person 3	P0	Task: API route missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
P09	Reject API exists	Person 3	P0	Task: API route missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
P10	Correction API exists	Person 3	P0	Task: API route missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
Q04	Extractions link to documents	Person 3	P0	Task: DB relationship missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
Q05	Validations link to documents	Person 3	P0	Task: DB relationship missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
Q06	Validations link to rule IDs	Person 3	P0	Task: DB column missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
Q07	Reviews link to documents	Person 3	P0	Task: DB relationship missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
Q08	Reviews link to records	Person 3	P0	Task: DB relationship missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
S02	OCR parser tests exist	Person 3	P1	Task: Test file missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail

S03	Field extraction tests exist	Person 3	P1	Task: Test file missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
S04	Validation tests exist	Person 3	P1	Task: Test file missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail
S05	Review workflow tests exist	Person 3	P1	Task: Test file missing UI/UX: Dedicated /review page, side-by-side verification drawer & audit log detail